
Predicting Customer Churn at SyriaTel Using Machine Learning

A Strategic Initiative for Customer Retention and Revenue Protection.

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EXECUTIVE BRIEFING

SYRIATEL COMMUNICATIONS

Business Understanding

Key Stakeholder

SyriaTel marketing teams

Strategic Objectives

SyriaTel is experiencing customer churn, leading to **revenue loss** and increased costs associated with **acquiring new customers**.

The goal is to develop a **predictive model** that identifies at-risk customers early.



Targeted marketing



Improved customer service



Personalized retention strategies

A key focus is **minimizing missed churners**, as failing to identify these customers represents **lost revenue**.

Data Understanding

THE CHURN REALITY FROM
SYRIALTEL HISTORICAL
DATA

14%

Average Churn Rate

3,333

Total Records Analyzed

Zero

Missing Values Found
Zero duplicates

Class

Imbalance Noted

The analysis is based on **3,333 records** with **no missing values and duplicats**, ensuring a high-quality data foundation. Features capture both **behavioral** and **service-related** drivers.

Data Preparation

1

Step 1: Cleaning

Removed irrelevant features such as **phone number** and **state** to focus on predictive signals.

2

Step 2: Encoding

Encoded **categorical variables** into numeric format to ensure compatibility with machine learning algorithms.

3

Step 3: Splitting

Split data into **training and testing sets** to allow for unbiased model evaluation.

4

Step 4: Scaling

Applied **feature scaling** for Logistic Regression and prevented **data leakage** by fitting transformations only on training data.

Modeling Approach

01

Stage 1: Baseline

Logistic Regression

Simple and interpretable. High accuracy but **poor churn detection**.

02

Stage 2: Improved

Decision Tree

Captures **non-linear relationships**.
Significantly improved **recall**.

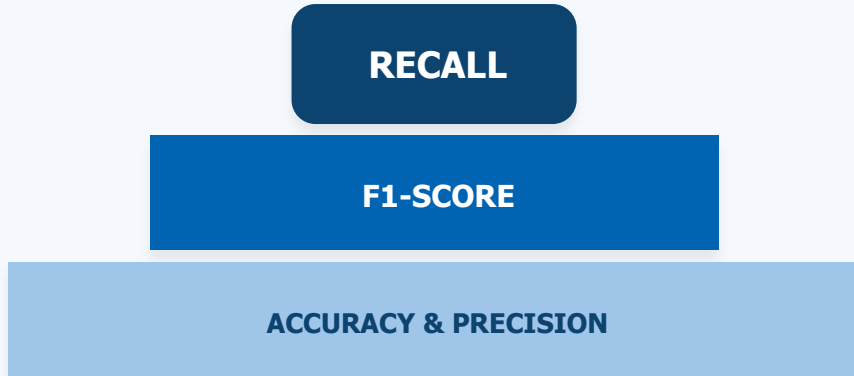
03

Stage 3: Refined

Tuned Decision Tree

Reduced complexity via **hyperparameter tuning**. Controlled for overfitting.

Evaluation



Hierarchy of Business Importance

Why Recall Matters

Recall is our **primary metric** for business decision-making.

Missing a churner means **losing a customer** without intervention, which **directly impacts revenue**.

The **Confusion Matrix** was utilized to visualize prediction errors across Accuracy, Precision, and F1-score.

Logistic Regression

Benchmark Performance

This model established the **benchmark** and why it was useful for comparison.

It emphasizes that although the baseline achieved **high accuracy**, it performed **poorly at identifying churners**.

BASELINE RECALL

0.18

⚠️ Poor Churn Identification

IMPROVED RECALL

0.73

 Significant Detection Gain

Decision Tree

Capturing Complexity

This model improved the **churn detection objective**.

It highlights the **Decision Tree's ability** to capture **non-linear relationships** and provide a better balance between **precision and recall**.

Tuned Decision Tree

Complexity Management

Refinement effort used to **manage model complexity** and **reduce overfitting**.

It makes clear that tuning did not meaningfully improve **recall** beyond the base Decision Tree.



Model Stability

Overfitting Protection Achieved

0.92 Accuracy Maintained

Model Comparison

Comparative Performance Metrics

Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	0.86	0.60	0.18	0.27
Decision Tree	0.92	0.73	0.73	0.73
Tuned Decision Tree	0.92	0.73	0.71	0.72

Key Insights



Service Friction

Customers with **frequent customer service calls** are significantly more likely to churn.



Usage Intensity

High usage (minutes and charges) is associated with an **increased churn risk**.



Dissatisfaction

Customer dissatisfaction remains a major driver of churn across the SyriaTel dataset.

Conclusion

The Decision Tree model outperformed the baseline, correctly identifying 73% of churners.

This provides the best balance between **performance** and **business value** for SyriaTel's retention strategy.

Business Recommendations



Risk Identification

Use the **Decision Tree model** to identify customers at **high risk of churn** before they leave the network.



Retention Tactics

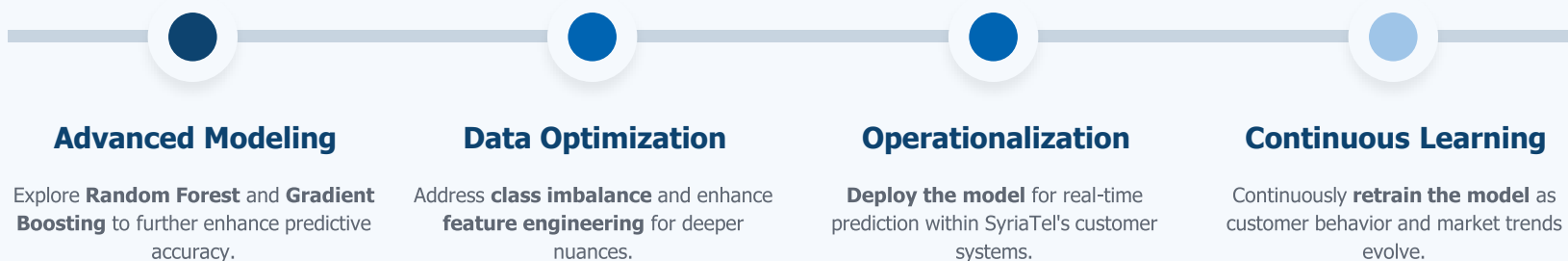
- **Personalized offers**
- **Proactive customer support**
- **Service or pricing adjustments**

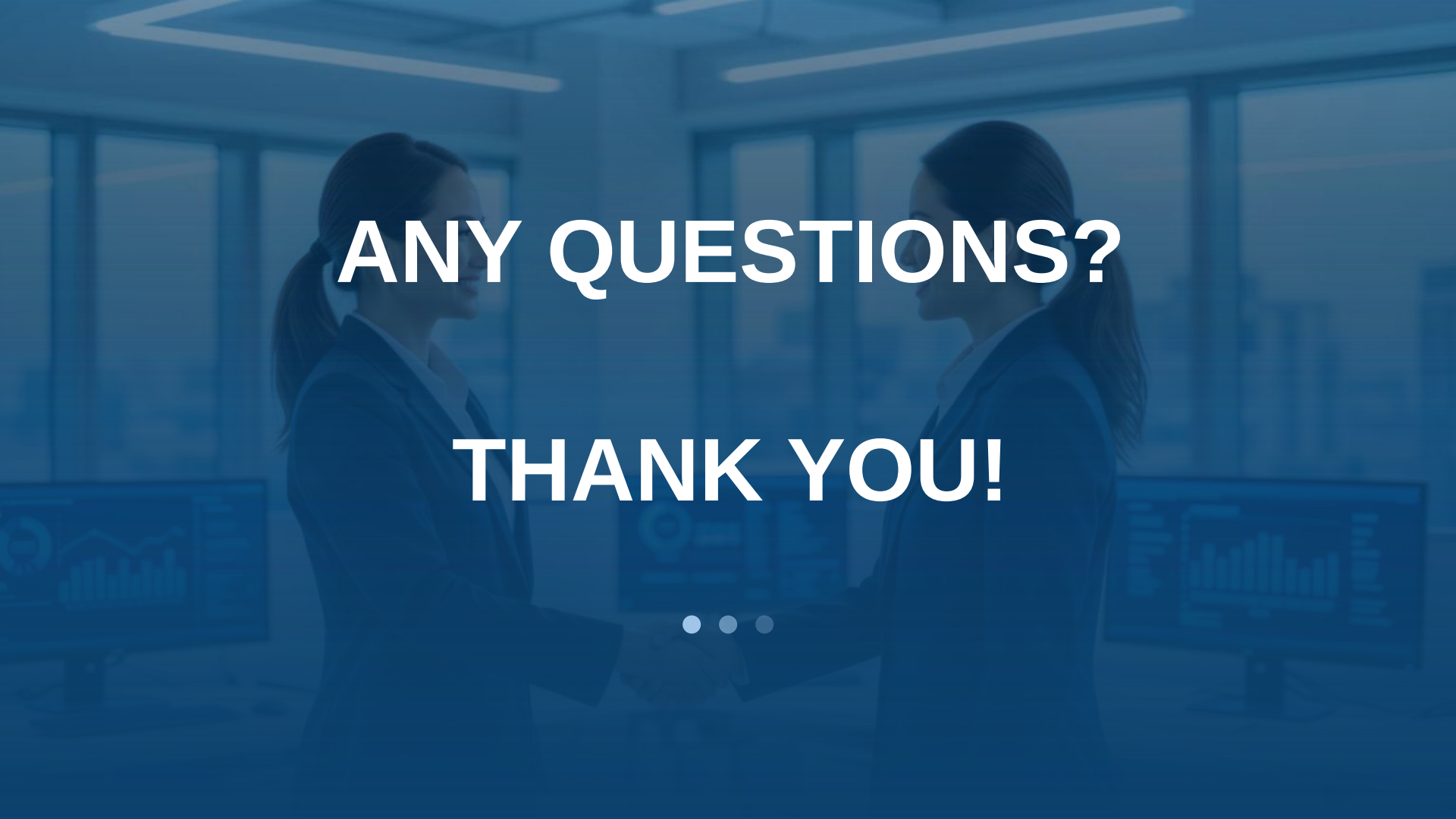


Revenue Protection

Focus on **early identification** to reduce customer attrition and **protect corporate revenue**.

Next Steps



A blue-tinted photograph of two businesswomen in an office setting, shaking hands. They are both wearing blouses and have their hair in ponytails. In the background, there are computer monitors displaying charts and graphs, and large windows letting in light. The overall mood is professional and positive.

ANY QUESTIONS?

THANK YOU!

